

SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022**Mathematics****MAT 6B 13 (E02)—LINEAR PROGRAMMING****(2014 to 2018 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 13 (E02)—LINEAR PROGRAMMING

(Multiple Choice Questions for SDE Candidates)

1. For any two points x and y in $\mathbb{R}^n$, the set $\{u : u = \lambda x + (1 - \lambda)y, 0 \leq \lambda \leq 1\}$ is called :
 - (A) The circle.
 - (B) A parabola.
 - (C) An ellipse.
 - (D) The line segment joining the points x and y .
2. The intersection of a finite number of convex sets is :
 - (A) Not convex.
 - (B) A half space.
 - (C) A cone.
 - (D) Convex.
3. A set that contains the ϵ -nbd of each of its points, is called :
 - (A) The boundary of the set.
 - (B) The closure of the set.
 - (C) An open set.
 - (D) A closed set.
4. A set is said to be closed if its, :
 - (A) Complement is open.
 - (B) Complement is not open.
 - (C) Complement is convex.
 - (D) Complement is not convex.
5. Let S be a convex subset of the plane, bounded by lines in the plane. Then a linear function $z = c_1x_1 + c_2x_2$, where $x_1, x_2 \in S$, c_1 and c_2 are scalars, attains its extreme values at :
 - (A) The interior of S .
 - (B) The vertices of S .
 - (C) The exterior of S .
 - (D) The X-axis.
6. In the iteration of simplex method, if $z_j - c_j \geq 0$ for all j , then the initial basic feasible solution is :
 - (A) Not a solution.
 - (B) Not optimal.
 - (C) An optimum solution.
 - (D) None of the above.
7. A feasible solution to an L.P.P. which is also a basic solution to the problem is called :
 - (A) An optimum solution to the L.P.P.
 - (B) A standard solution to the L.P.P.
 - (C) A basic feasible solution to the L.P.P.
 - (D) A feasible solution to the L.P.P.
8. If the number of dual variables are m and primal constraints are n , then :
 - (A) $m \neq n$.
 - (B) $m > n$.
 - (C) $m < n$.
 - (D) $m = n$.

9. The dual of dual problem is :
- (A) The unsymmetric dual problem. (B) The unsymmetric primal problem.
(C) The dual problem. (D) The primal problem.
10. A balanced transportation problem has :
- (A) An optimal solution always. (B) No solution.
(C) No feasible solution. (D) No optimal solution.
11. A system of n linear equations $\mathbf{Ax} = \mathbf{b}$ is called a triangular system if the matrix $\mathbf{A}$ is :
- (A) Unit matrix. (B) Zero matrix.
(C) Diagonal matrix. (D) Triangular matrix.
12. An initial basic feasible solution to a T.P. is obtained by :
- (A) Method of penalties. (B) Two-phase simplex method.
(C) Row minima method. (D) Big M method.
13. An L.P.P. can be solve using graphical method if it has :
- (A) More than four variables. (B) Only two variables.
(C) More than two variables. (D) Three variables.
14. The assignment problem is a special case of transportation problem because :
- (A) The number of origins is not equal to number of destinations.
(B) The number of origins is greater than number of destinations.
(C) The number of origins is less than number of destinations.
(D) The number of origins is equal to number of destinations.
15. $Cl(2, 7) = \underline{\hspace{2cm}}$.
- (A) $(2, 7)$. (B) $(2, 7]$.
(C) $[2, 7)$. (D) $[2, 7]$.
16. A simplex in zero dimension is :
- (A) A point. (B) A line.
(C) A square. (D) None of the above.
17. A simplex in one dimension is :
- (A) A point. (B) A line segment.
(C) A square. (D) None of the above.

18. The optimal solution to an L.P.P. is :

- (A) Always infinite.
- (B) Always finite.
- (C) Unique.
- (D) Either unique or infinite.

19. The simplex method is :

- (A) An iterative method.
- (B) A direct method.
- (C) Both direct and iterative method.
- (D) None of the above.

20. Loop is associated with :

- (A) Two-phase simplex method.
- (B) Big M method.
- (C) Assignment problem.
- (D) Transportation problem.

SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022

Mathematics

MAT 6B 13 (E02)—LINEAR PROGRAMMING

(2014 to 2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Section A

*Answer all questions.
Each question carries 1 mark.*

1. Define vertex of a convex set.
2. Find the convex hull of $\{x_1, x_2\} \subset \mathbb{R}^2$.
3. Define objective function.
4. While solving an LPP by graphical method, when does one conclude that there exist infinitely many points in the feasible region at which the objective function attains optimum ?
5. Define a slack variable.
6. Write the matrix form of general LPP.
7. What is meant by feasible solution to an LPP ?
8. What is meant by optimal solution to an LPP ?
9. When do you say that a transportation problem is unbalanced ?
10. State the necessary and sufficient condition for the existence of a feasible solution to general transportation problem.
11. Write the number of basic variables of the general transportation problem at any stage of feasible solution.
12. In assignment problem, what is the value of decision variable ?

(12 × 1 = 12 marks)

Section B

*Answer any nine questions.
Each question carries 2 marks.*

13. Prove that the intersection of two convex sets is a convex set.
14. Write the standard form of LPP.
15. What is the basic principle of linear programming ?

Turn over

16. Does a feasible solution exist for the LPP : Maximize $z = 2x_1 + 10x_2$ subject to $x_1 - x_2 \geq 1, -x_1 + x_2 \geq 2, x_1 \geq 0, x_2 \geq 0$? Give reasons.
17. For linear inequalities, show that the solution set for a group of inequalities is a convex set.
18. Write the dual of the LPP :
Maximize $f(x) = 3x_1 + 2x_2$ subject to $2x_1 + x_2 \leq 20, x_1 + 3x_2 \leq 20, x_1 \geq 0, x_2 \geq 0$.
19. Define a loop in transportation table.
20. How does a loop in a transportation table related to a basic feasible solution ?
21. Define a triangular basis. Write the role of triangular basis in transportation problem.
22. How do you solve an unbalanced transportation problem ?
23. Write the disadvantage of North-West corner rule.
24. "An assignment problem is a particular case of a transportation problem." Justify.

(9 × 2 = 18 marks)

Section C

*Answer any six questions.
Each question carries 5 marks.*

25. Show that the set $S = \{(x_1, x_2) : 5x_1 + 2x_2 \geq 10, 2x_1 + 5x_2 \geq 10\}$ is convex.
26. (a) Define convex linear combination of a finite set of vectors.
(b) Show that the set of all convex linear combinations of a finite number of vectors $u_1, u_2, u_3, \dots, u_k \in \mathbb{R}^n$ is a convex set.
27. Prove that the set of feasible solutions to an LPP is a convex set.
28. Show that the following system of linear equations has a degenerate solution :
 $2x_1 + x_2 - x_3 = 2, 3x_1 + 2x_2 + x_3 = 3$.
29. Let $x_1 = 2, x_2 = 4$ and $x_3 = 1$ be a feasible solution to the system of equations $2x_1 - x_2 + 2x_3 = 2, x_1 + 4x_2 = 18$. Reduce this feasible solution to a basic feasible solution.
30. Use graphical method to solve the following LPP :
Maximize $z = 6x_1 + x_2$ subject to $2x_1 + x_2 \geq 3, x_2 - x_1 \geq 0, x_1 \geq 0, x_2 \geq 0$.
31. Prove that the dual of the dual is the primal.
32. How do you resolve the problem of degeneracy in transportation problem ?

33. Obtain an initial basic feasible solution to the following transportation problem using North-West Corner rule :

	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	5	3	6	2	19
O ₂	4	7	9	1	37
O ₃	3	4	7	5	34
Demand	16	18	31	25	

(6 × 5 = 30 marks)

Section D

Answer any **two** questions.
Each question carries 10 marks.

34. Use Simplex method to solve the LPP :

Maximize $Z = 4x_1 + 10x_2$ subject to

$$2x_1 + x_2 \leq 50$$

$$2x_1 + 5x_2 \leq 100$$

$$2x_1 + 3x_2 \leq 90 \text{ and } x_1 \geq 0, x_2 \geq 0.$$

35. Use Vogel's approximation method to obtain an initial basic feasible solution to the transportation problem :

	D ₁	D ₂	D ₃	D ₄	Availability
O ₁	11	13	17	14	250
O ₂	16	18	14	10	300
O ₃	21	24	13	10	400
Demand	200	225	275	250	

36. A department head has four subordinates and four tasks to be performed. The subordinates differ in efficiency and the tasks differ in their intrinsic difficulty. His estimate of the time each man would take to perform each task is given in the matrix below :

Tasks	Men			
	T ₁	T ₂	T ₃	T ₄
A	18	26	17	11
B	13	28	14	26
C	38	19	18	15
D	19	26	24	10

How the task should be allocated, one task to a man, so as to minimize the total man-hours ?

(2 × 10 = 20 marks)